

## AI ASSISTED RESEARCH PUBLICATION MANAGEMENT SYSTEM CREATED BY: SAHIL KUMAR SONI 24BCE10085

The AI-Assisted Research Publication Management System was successfully developed using Spring Boot and MongoDB. The application provides RESTful APIs for managing departments, researchers, publications, journals, conferences, and research profiles. CRUD operations, validation mechanisms, MongoDB integration, and relationship mappings including One-to-One, One-to-Many, Many-to-One, and Many-to-Many were successfully implemented and tested using Swagger UI. The system also includes scholar dashboard, publication search, and researcher search functionalities, making it a practical backend solution for research publication management.

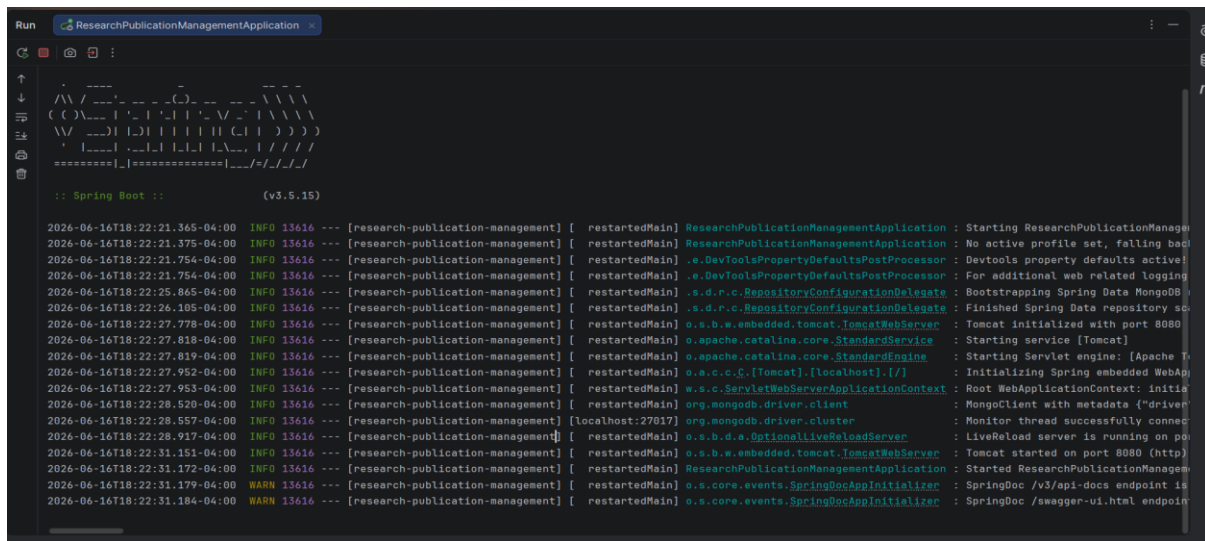
### Technologies Used

- Java 17
- Spring Boot
- Spring Data MongoDB
- MongoDB Compass
- Swagger OpenAPI
- Maven
- REST APIs

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Figure 0: Successful startup of Spring Boot application and MongoDB database connection.



```
Run ResearchPublicationManagementApplication x
...
:: Spring Boot ::                (v3.5.15)

2026-06-16T18:22:21.365-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] ResearchPublicationManagementApplication : Starting ResearchPublicationManagementApplication
2026-06-16T18:22:21.375-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] ResearchPublicationManagementApplication : No active profile set, falling back to default profiles
2026-06-16T18:22:21.754-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] o.s.devtools.properties.DevToolsPropertyDefaultsPostProcessor : DevTools property defaults activated
2026-06-16T18:22:21.754-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] o.s.devtools.properties.DevToolsPropertyDefaultsPostProcessor : For additional web related logging
2026-06-16T18:22:25.865-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] s.d.r.c.RepositoryConfigurationDelegate : Bootstrapping Spring Data MongoDB
2026-06-16T18:22:26.105-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] s.d.r.c.RepositoryConfigurationDelegate : Finished Spring Data repository scanning
2026-06-16T18:22:27.779-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat initialized with port 8080
2026-06-16T18:22:27.818-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-06-16T18:22:27.819-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] o.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.90]
2026-06-16T18:22:27.953-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Initializing Spring embedded WebApplicationContext
2026-06-16T18:22:27.953-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed
2026-06-16T18:22:28.520-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] org.mongodb.driver.client : MongoClient with metadata ("driver"
2026-06-16T18:22:28.557-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] org.mongodb.driver.cluster : Monitor thread successfully connected to server
2026-06-16T18:22:28.917-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] o.s.b.d.o.OptionalLiveReloadServer : LiveReload server is running on port 35729
2026-06-16T18:22:31.151-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port 8080 (http)
2026-06-16T18:22:31.172-04:00 INFO 13616 --- [research-publication-management] [ restartedMain] ResearchPublicationManagementApplication : Started ResearchPublicationManagementApplication
2026-06-16T18:22:31.179-04:00 WARN 13616 --- [research-publication-management] [ restartedMain] o.s.core.events.SpringDocAppInitializer : SpringDoc /v3/api-docs endpoint is
2026-06-16T18:22:31.184-04:00 WARN 13616 --- [research-publication-management] [ restartedMain] o.s.core.events.SpringDocAppInitializer : SpringDoc /swagger-ui.html endpoint is
```

Figure 1: Spring Boot project structure implementing layered architecture.

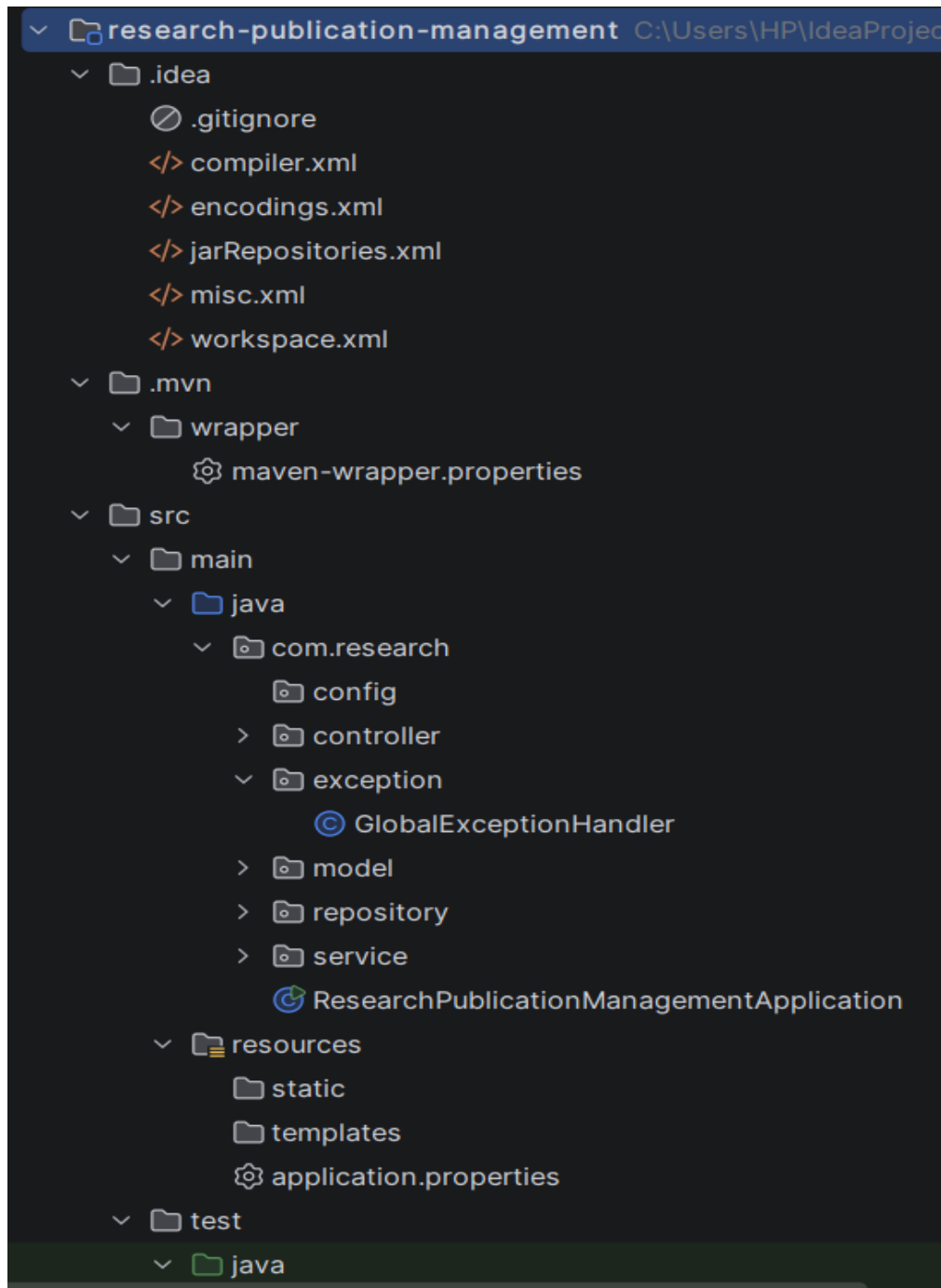
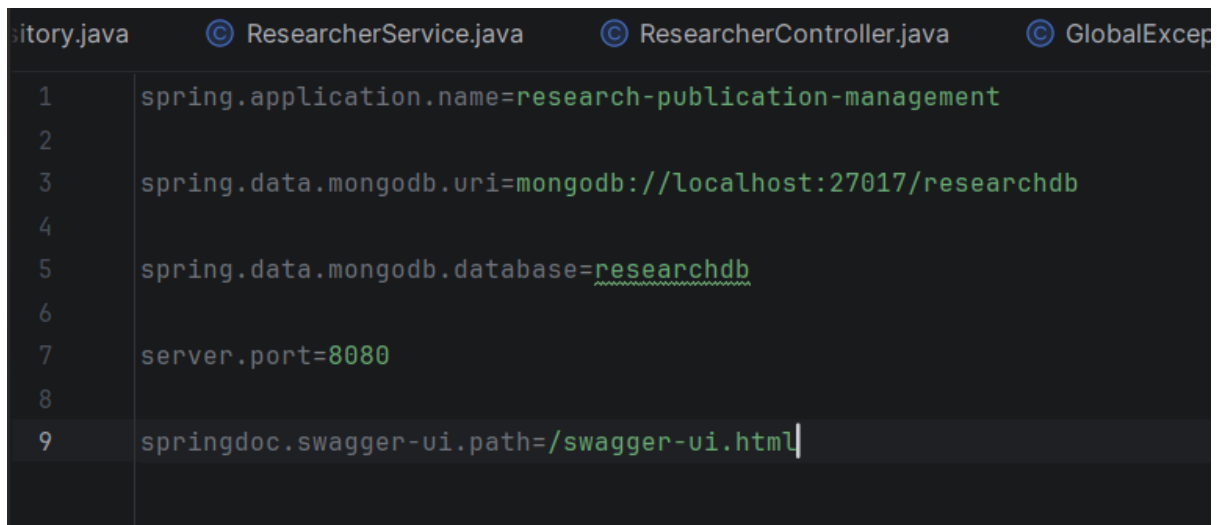
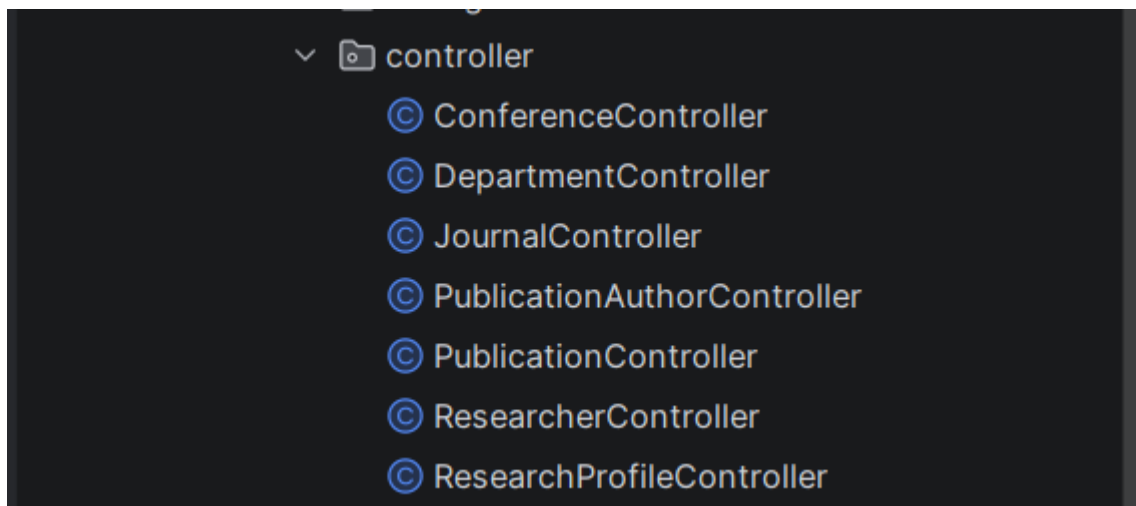


Figure 2: MongoDB database configuration.



```
History.java  © ResearcherService.java  © ResearcherController.java  © GlobalExcep
1  spring.application.name=research-publication-management
2
3  spring.data.mongodb.uri=mongodb://localhost:27017/researchdb
4
5  spring.data.mongodb.database=researchdb
6
7  server.port=8080
8
9  springdoc.swagger-ui.path=/swagger-ui.html
```

Figure 3: REST APIs exposed through Swagger UI.

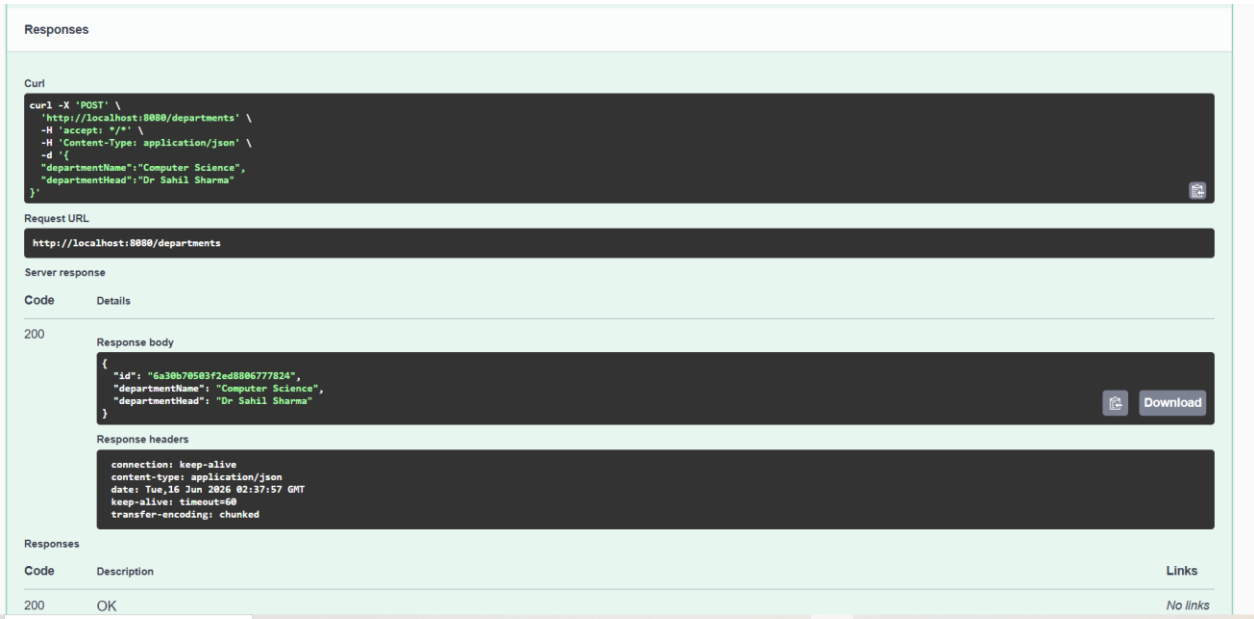
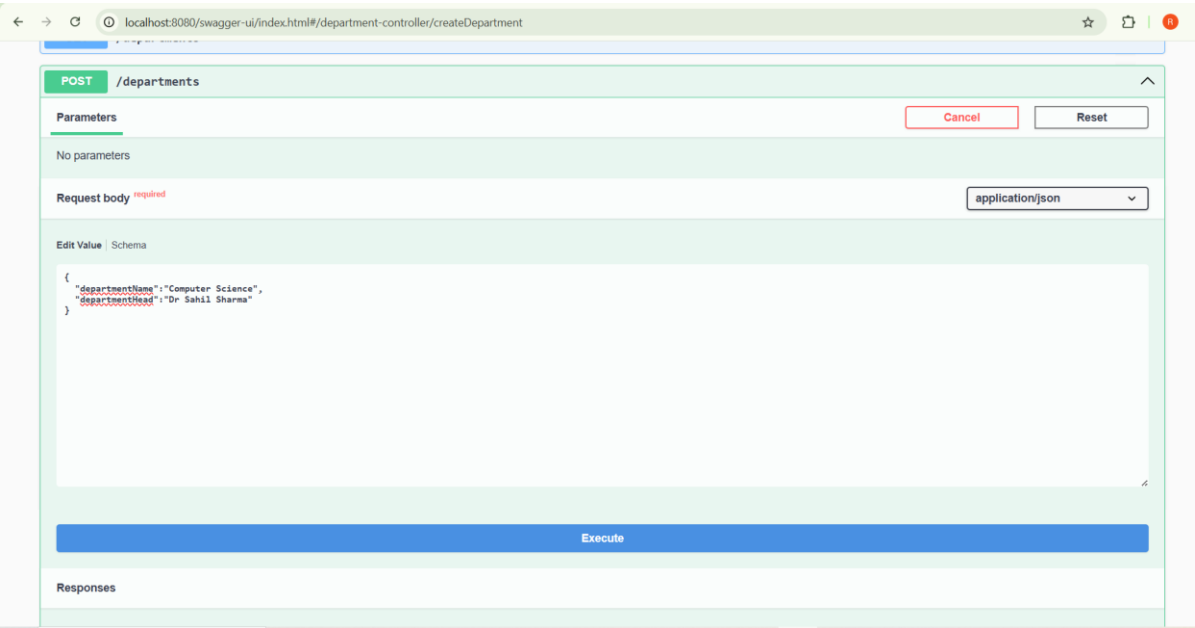


```

  controller
    © ConferenceController
    © DepartmentController
    © JournalController
    © PublicationAuthorController
    © PublicationController
    © ResearcherController
    © ResearchProfileController
```

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Figure 4: Department creation using REST API.



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Figure 5: Input validation using Spring Validation.

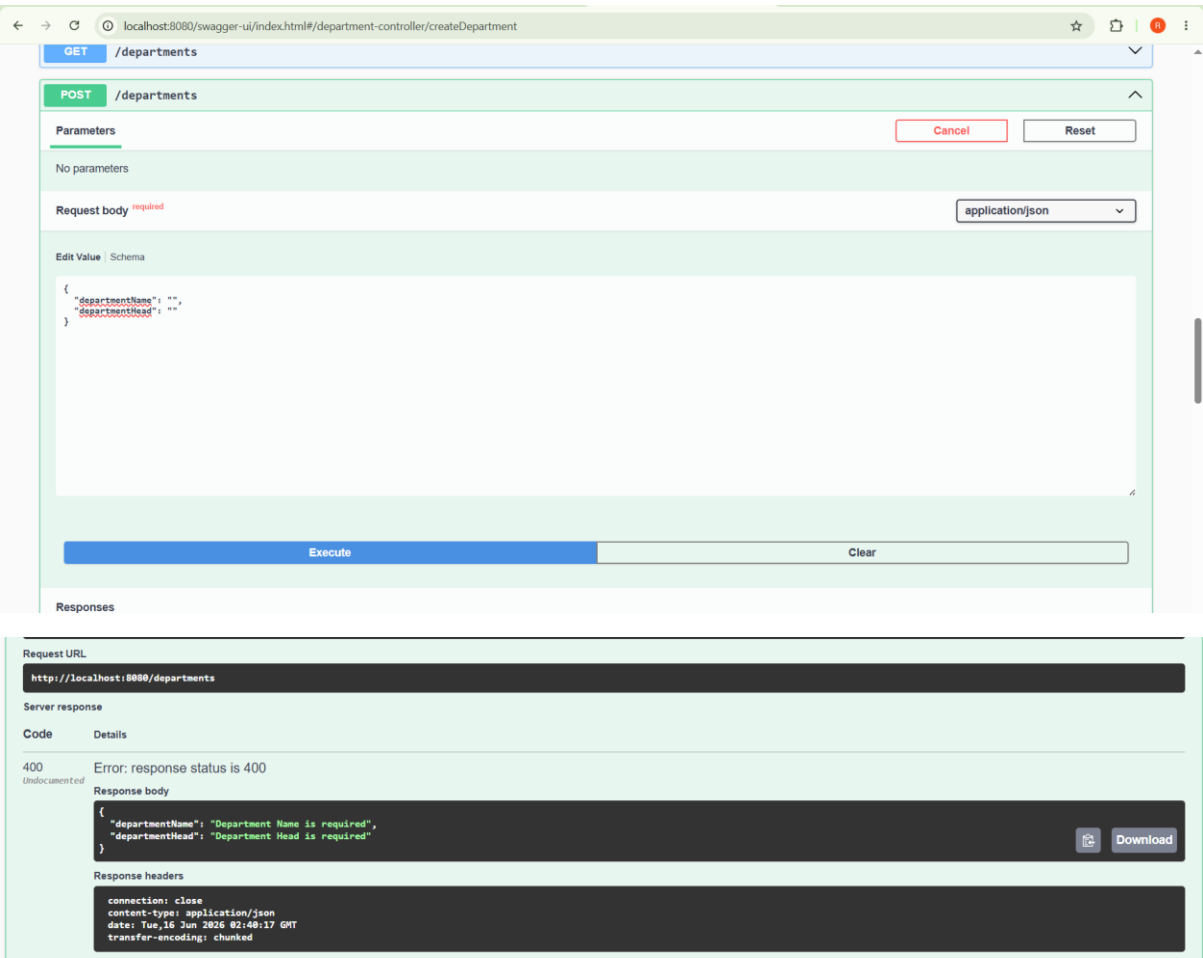
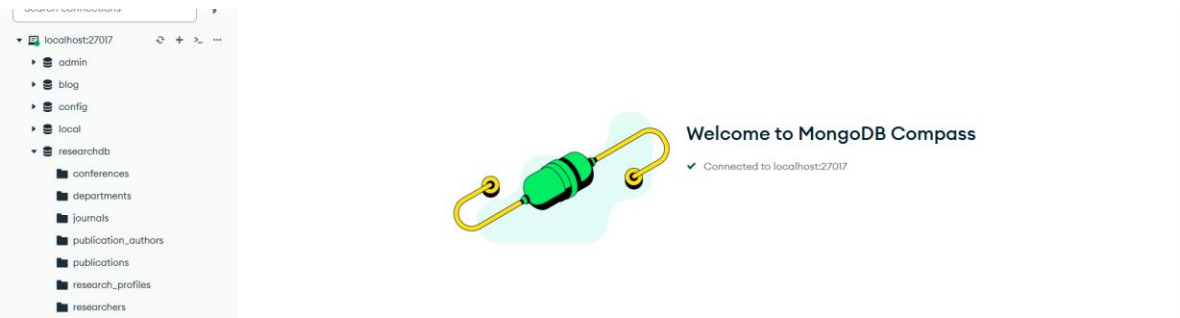
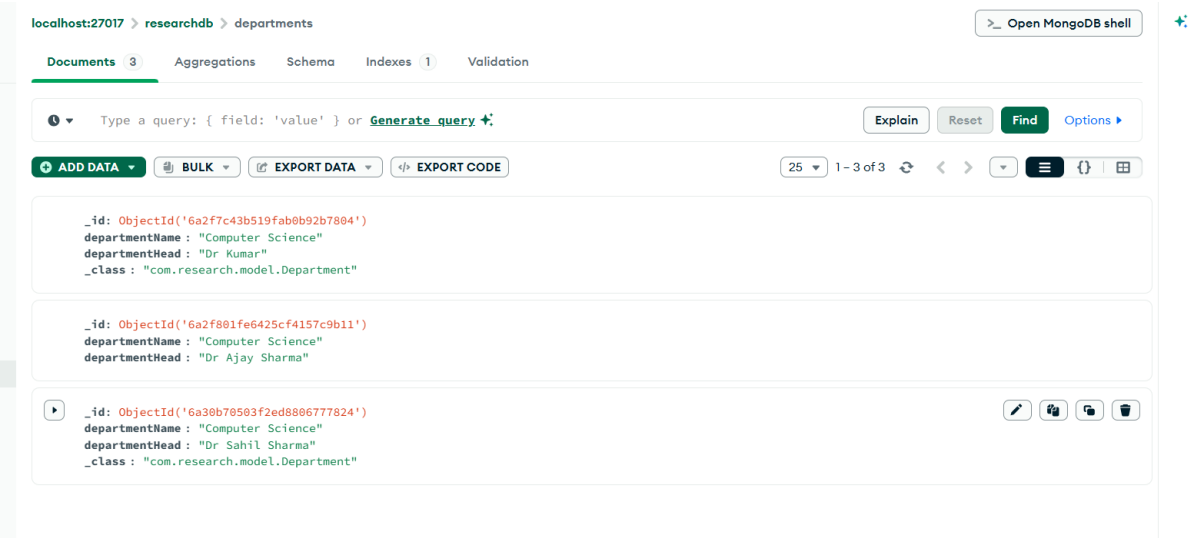


Figure 6: MongoDB collections created by the application.



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Figure 7: Department records stored in MongoDB.

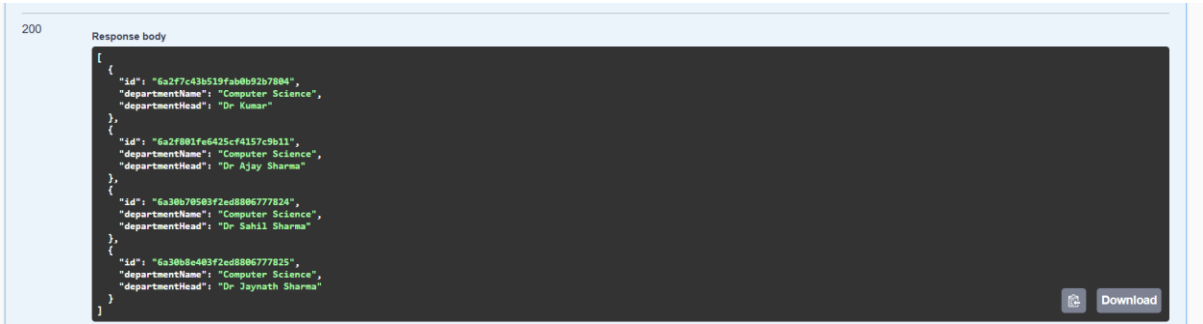


CRUD OPERATION:

CREATE:



READ:



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UPDATE:

200

Response body

```
{
  "id": "6a2f7c43b519fab8b92b7884",
  "departmentName": "Computer Science",
  "departmentHead": "Dr Sachin Kumar"
}
```

Download

Response headers

```
connection: keep-alive
content-type: application/json
date: Tue, 16 Jun 2026 02:49:05 GMT
keep-alive: timeout=60
transfer-encoding: chunked
```

Responses

DELETE:

200

Response body

Department Deleted Successfully

Download

Response headers

```
connection: keep-alive
content-length: 31
content-type: text/plain; charset=UTF-8
date: Tue, 16 Jun 2026 02:50:30 GMT
keep-alive: timeout=60
```

Responses

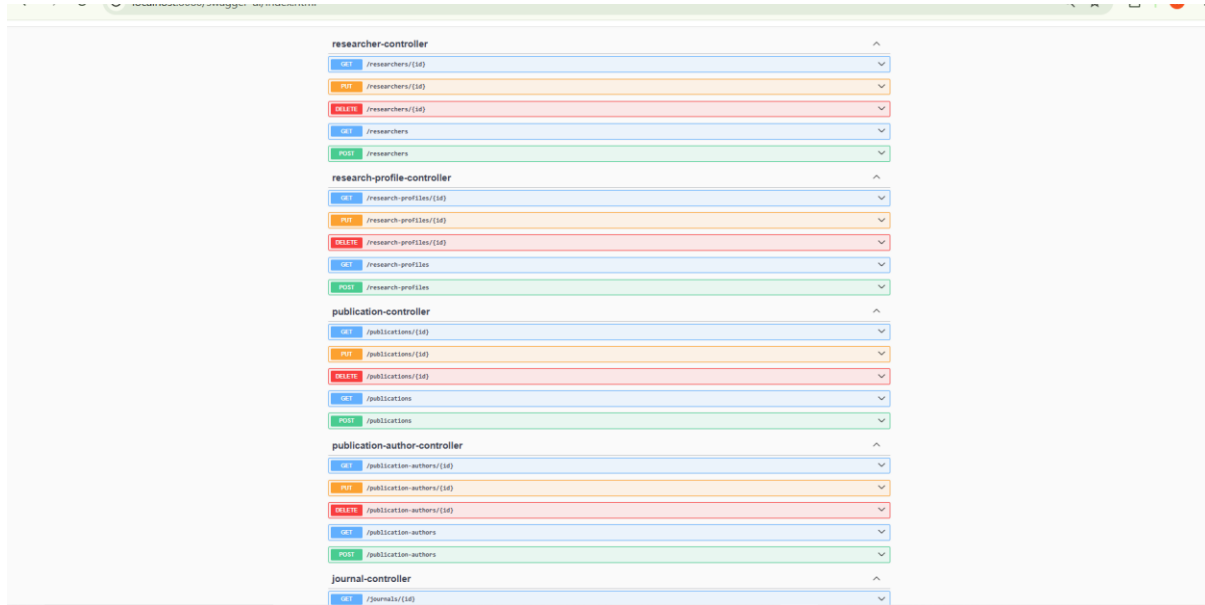
| Code | Description | Links |
|------|-------------|-------|
|------|-------------|-------|

MONGODB DEPARTMENT

|                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>_id: ObjectId('6a2f7c43b519fab8b92b7884') departmentName: "Computer Science" departmentHead: "Dr Kumar" _class: "com.research.model.Department"</pre>        |
| <pre>_id: ObjectId('6a2f801fe6425cf4157c9b11') departmentName: "Computer Science" departmentHead: "Dr Ajay Sharma"</pre>                                          |
| <pre>_id: ObjectId('6a30b70503f2ed8806777824') departmentName: "Computer Science" departmentHead: "Dr Sahil Sharma" _class: "com.research.model.Department"</pre> |

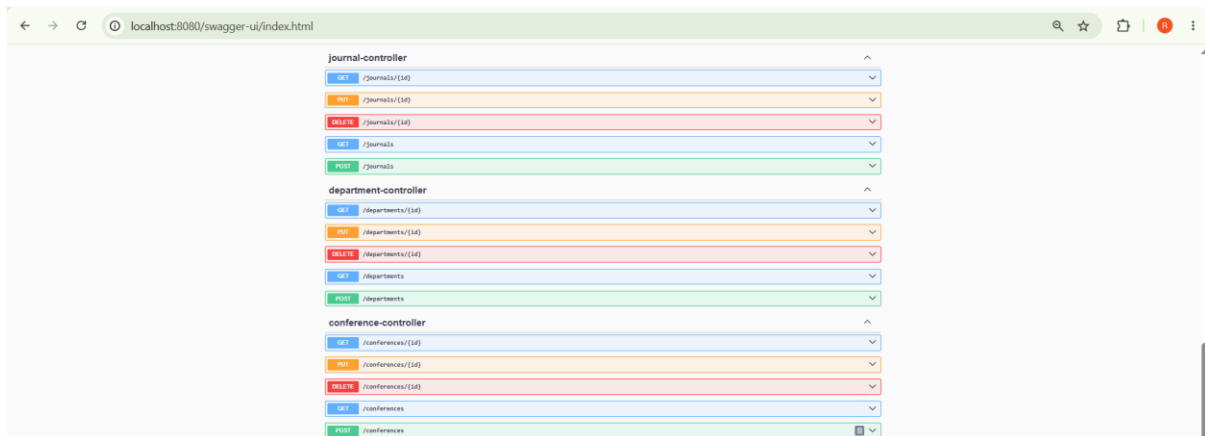
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## SCHEMA :



This screenshot displays the Swagger UI for the AI Assisted Research Publication Management System. It lists the following API endpoints:

| Controller                    | Method | Endpoint                  |
|-------------------------------|--------|---------------------------|
| researcher-controller         | GET    | /researchers/{id}         |
|                               | POST   | /researchers/{id}         |
|                               | DELETE | /researchers/{id}         |
|                               | GET    | /researchers              |
|                               | POST   | /researchers              |
| research-profile-controller   | GET    | /research-profiles/{id}   |
|                               | POST   | /research-profiles/{id}   |
|                               | DELETE | /research-profiles/{id}   |
|                               | GET    | /research-profiles        |
|                               | POST   | /research-profiles        |
| publication-controller        | GET    | /publications/{id}        |
|                               | POST   | /publications/{id}        |
|                               | DELETE | /publications/{id}        |
|                               | GET    | /publications             |
|                               | POST   | /publications             |
| publication-author-controller | GET    | /publication-authors/{id} |
|                               | POST   | /publication-authors/{id} |
|                               | DELETE | /publication-authors/{id} |
|                               | GET    | /publication-authors      |
|                               | POST   | /publication-authors      |
| journal-controller            | GET    | /journals/{id}            |
|                               | POST   | /journals/{id}            |



This screenshot displays the Swagger UI for the AI Assisted Research Publication Management System, showing additional API endpoints:

| Controller            | Method | Endpoint          |
|-----------------------|--------|-------------------|
| journal-controller    | GET    | /journals/{id}    |
|                       | POST   | /journals/{id}    |
|                       | DELETE | /journals/{id}    |
|                       | GET    | /journals         |
|                       | POST   | /journals         |
| department-controller | GET    | /departments/{id} |
|                       | POST   | /departments/{id} |
|                       | DELETE | /departments/{id} |
|                       | GET    | /departments      |
|                       | POST   | /departments      |
| conference-controller | GET    | /conferences/{id} |
|                       | POST   | /conferences/{id} |
|                       | DELETE | /conferences/{id} |
|                       | GET    | /conferences      |
|                       | POST   | /conferences      |

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FINAL SCHEMA:



Fig: Scholar Dashboard displaying real-time statistics, system status, date, time, and motivational research insights.

- Publication search functionality enabling keyword-based retrieval of research publications.
- Researcher search functionality for locating researchers based on their names.
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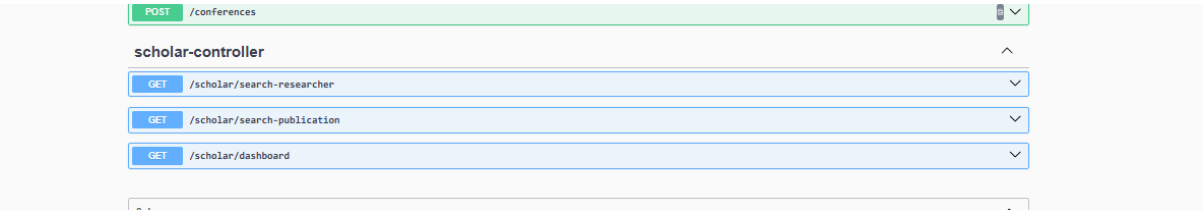
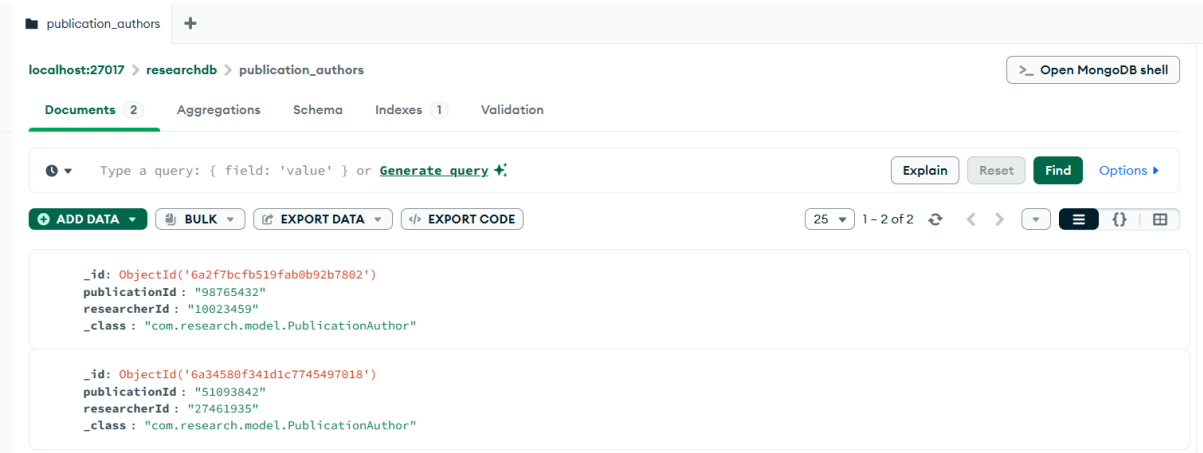
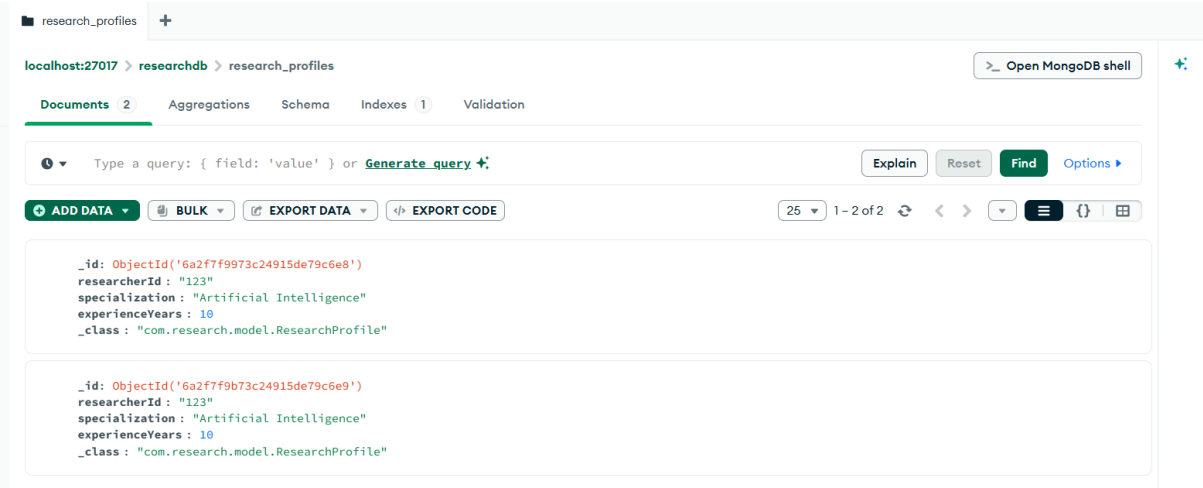


Figure : Many-to-Many relationship mapping between Researchers and Publications using the Publication Author collection.



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Figure : One-to-One relationship implementation between Researcher and Research Profile entities.



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Figure : Scholar Dashboard displaying real-time research statistics, system status, date, and time.

GET

/scholar/dashboard

Parameters

Cancel

No parameters

Execute

Clear

Responses

Curl

curl -X 'GET' \nhttp://localhost:8080/scholar/dashboard' \n-H 'accept: \*/\*' \n

Request URL

http://localhost:8080/scholar/dashboard

Server response

CodeDetails

200

Response body

{\n "currentDate": "2025-06-16",\n "currentTime": "12:45:46",\n "motivation": "The best way to predict the future is to invent it.",\n "totalDepartments": 3,\n "totalResearchers": 5,\n "totalPublications": 5,\n "totalJournals": 0,\n "totalConferences": 4,\n "status": "ACTIVE"\n}

Download

Response headers

connection: keep-alive\ncontent-type: application/json\ndate: Tue, 16 Jun 2025 12:45:46 GMT\nkeep-alive: timeout=60\ntransfer-encoding: chunked

Responses

CodeDescriptionLinks

200OK

Media type

\*/\*

Controls Accept header.

Example Value | Schema

{\n "currentDate": "string",\n "currentTime": "string",\n "motivation": "string",\n "totalDepartments": 0,\n "totalResearchers": 0,\n "totalPublications": 0,\n "totalJournals": 0,\n "totalConferences": 0,\n "status": "string"\n}

No links

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Figure : Publication search functionality enabling keyword-based retrieval of research publications.

GET

/scholar/search-publication

Parameters

Cancel

| Name                          | Description |
|-------------------------------|-------------|
| keyword <span>required</span> |             |
| string                        |             |
| (query)                       |             |

ExecuteClear

Responses

Curl

curl -X 'GET' \nhttp://localhost:8080/scholar/search-publication?keyword=AI' \n-H 'accept: \*/\*' \n

Request URL

http://localhost:8080/scholar/search-publication?keyword=AI

Server response

| Code | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div><div>Response body</div><div><pre>{\n  \"id\": \"6a2f7baab519fab6b2b7801\",\n  \"title\": \"AI Assisted Research Publication Management\",\n  \"publicationYear\": 2020,\n  \"conferenceId\": \"someConferenceId\"\n},\n  {\n    \"id\": \"6a30c278d02114078d0c3025\",\n    \"title\": \"Cyber Security using AI\",\n    \"publicationYear\": 2020,\n    \"conferenceId\": \"CONF003\"\n  }\n}</pre></div><div>Download</div></div> <div><div>Response headers</div><div><pre>connection: keep-alive\ncontent-type: application/json\ndate: Tue, 16 Jun 2026 22:48:42 GMT\nkeep-alive: timeout=60\ntransfer-encoding: chunked</pre></div></div> |

Responses

| Code | Description | Links |
|------|-------------|-------|
|------|-------------|-------|

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Figure : Researcher search functionality for locating researchers based on their names.

scholar-controller

GET /scholar/search-researcher

Parameters

Cancel

| Name            | Description |
|-----------------|-------------|
| name * required |             |
| string          |             |
| (query)         |             |

ExecuteClear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8080/scholar/search-researcher?name=Rahul' \
  -H 'accept: */*'
```

Request URL

```
http://localhost:8080/scholar/search-researcher?name=Rahul
```

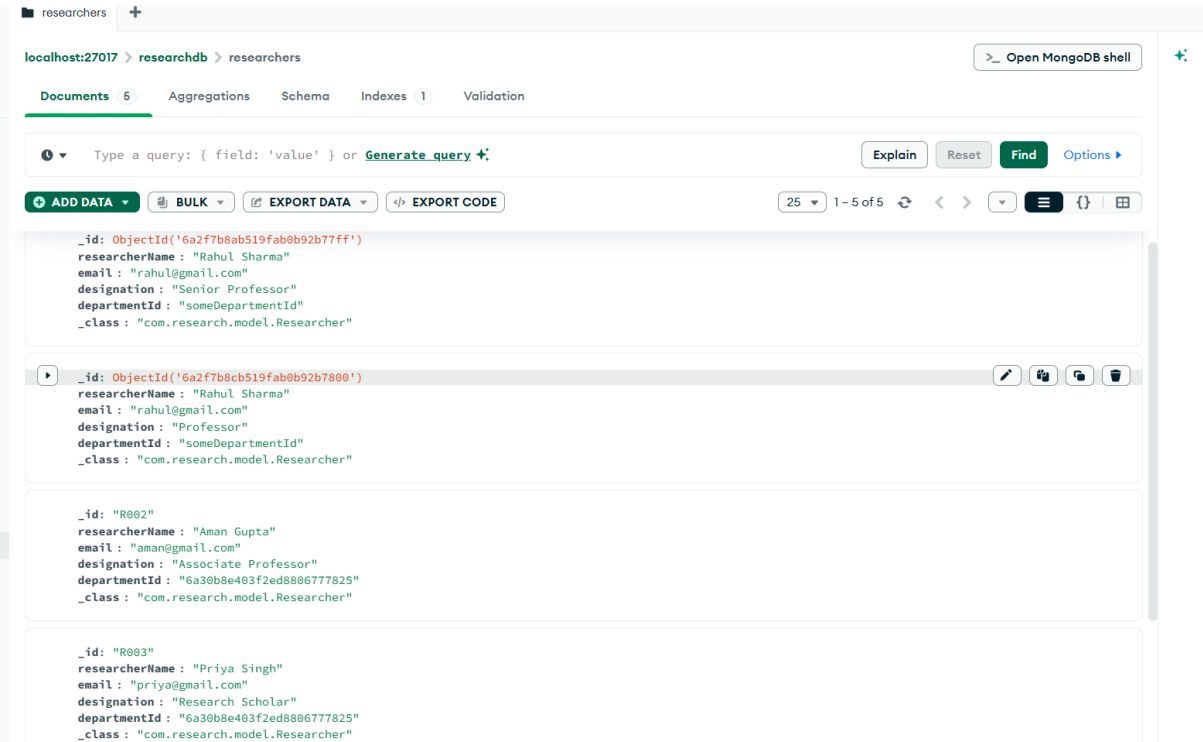
Server response

| Code | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 200  | <div><div>Response body</div><div><pre>{   "id": "6a2f7b8d519fab082b77ff",   "researcherName": "Rahul Sharma",   "email": "rahul@gmail.com",   "designation": "Senior Professor",   "departmentId": "someDepartmentId" }, {   "id": "6a2f7b8d519fab082b7800",   "researcherName": "Rahul Sharma",   "email": "rahul@gmail.com",   "designation": "Professor",   "departmentId": "someDepartmentId" }, {   "id": "6a2f802ee6425cf4157c9b12",   "researcherName": "Rahul Sharma",   "email": "rahul@gmail.com",   "designation": "Professor",   "departmentId": "CS001" } ]</pre></div><div>Download</div></div> <div>Response headers</div> <div><pre>connection: keep-alive content-type: application/json date: Tue, 16 Jun 2026 22:51:46 GMT keep-alive: timeout=60 transfer-encoding: chunked</pre></div> |

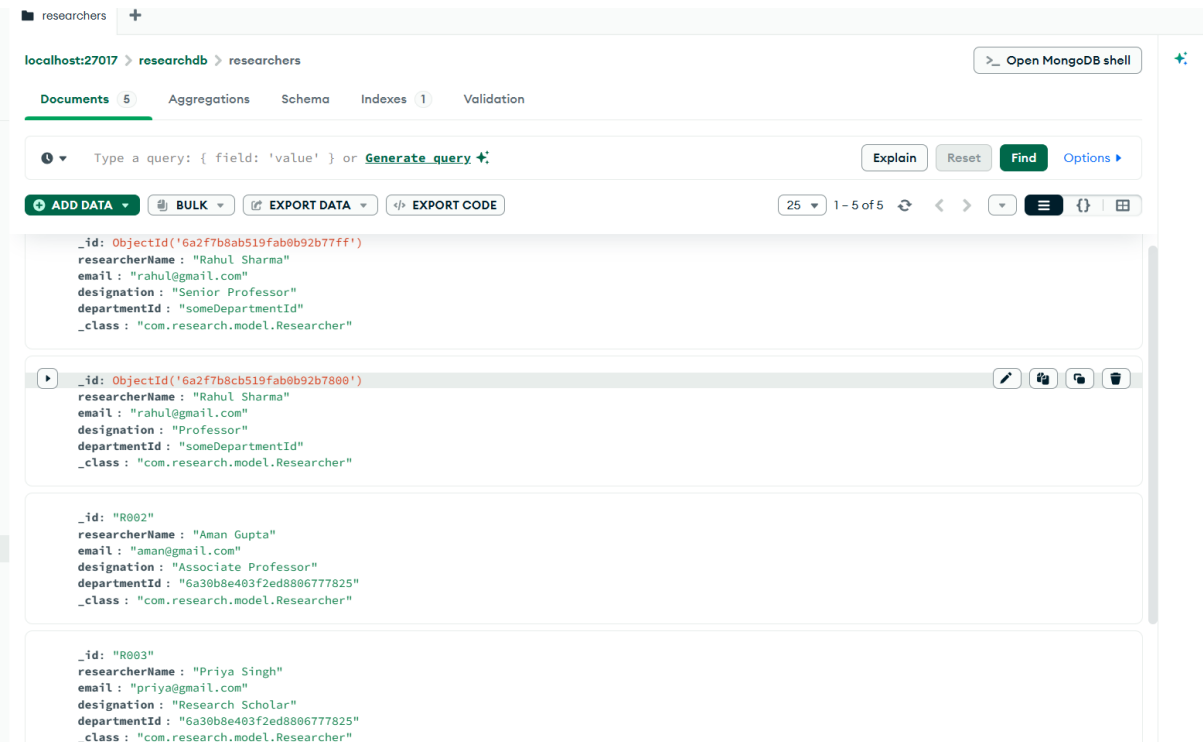
Responses

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Figure : One-to-Many relationship between Department and Researchers.



Many-to-One relationship between Researchers and Department, where multiple researchers are associated with a single department through the departmentId reference.



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Fig : Conference records stored in MongoDB for managing conference information and publication events.

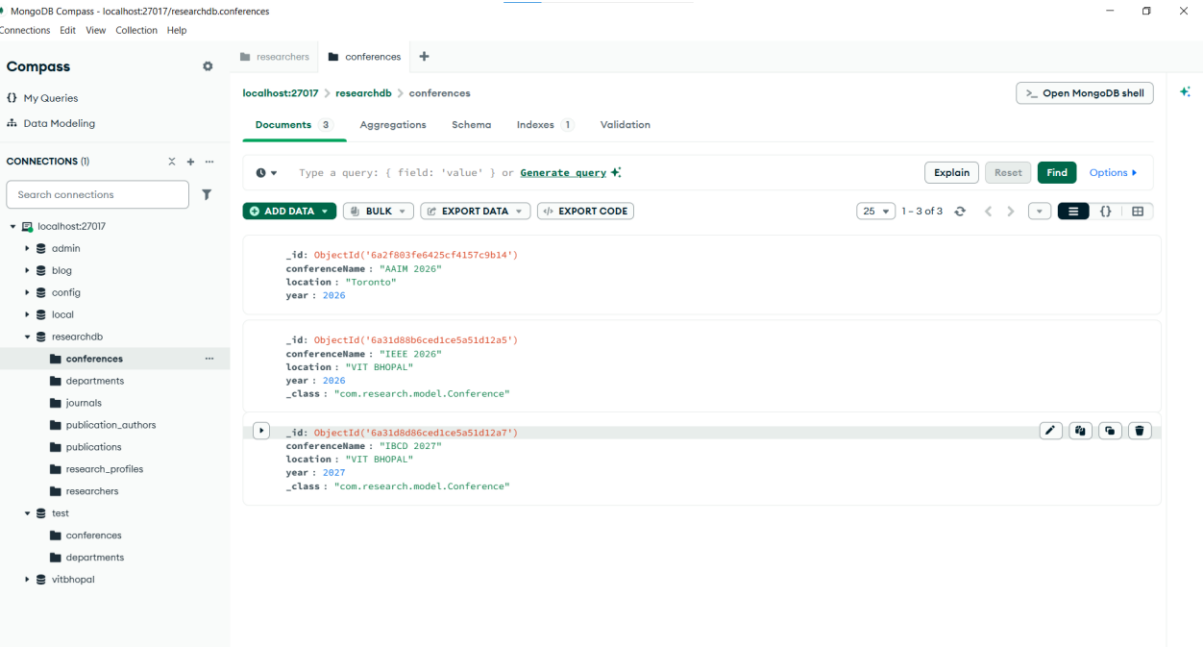


Fig: Journal records stored in MongoDB for publication categorization and management.

